

João Vitor Oliveira Rodrigues

Technology as a Survival Factor: Proposal for an Integrated Management System
to Reduce MEI Mortality

SÃO PAULO
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João Vitor Oliveira Rodrigues

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to Reduce MEI Mortality

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Advisor: Prof. Bruna Mayer

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Epigraph

“O sucesso é a soma de pequenos esforços repetidos dia após dia” (Robert Collier, s.d)

Resumo

Oliveira Rodrigues, João Vitor. **Tecnologia como fator de sobrevivência**. 2025. 42p. TCC (Graduação) – Curso Computer Engineering, Instituto de Tecnologia e Liderança, São Paulo, 2025.

Este trabalho possui caráter técnico e empreendedor, fundamentando o desenvolvimento e o plano de negócios da ViaFacilis, uma startup voltada à gestão de pequenos negócios. O projeto nasce da identificação de uma falha crítica no ecossistema brasileiro: a alta taxa de mortalidade dos Microempreendedores Individuais (MEIs), causada majoritariamente pela ineficiência de gestão. A iniciativa justifica-se pela premissa de que a tecnologia acessível é o fator determinante para a sobrevivência e escalabilidade dos empreendimentos no cenário atual. Com base nisso, o propósito da ViaFacilis é atuar diretamente na redução dessa métrica de mortalidade, oferecendo uma plataforma centralizada, modular e *offline-first*, visando a redução de processos manuais feitos com tecnologias dispersas. A metodologia percorreu quatro fases estratégicas: validação da problemática baseada em dados do SEBRAE, estruturação legal e análise de riscos, desenvolvimento técnico do Produto Mínimo Viável (MVP) e a elaboração da estratégia de *go-to-market*. Os

resultados obtidos com o MVP demonstram a capacidade da solução em otimizar o fluxo de caixa, o controle de estoque e a geração do DASN SIMEI, centralizando as principais demandas de gestão dos MEIs.

Palavras-Chave: empreendedorismo; gestão para MEI; startup; plano de negócios.

ABSTRACT

Oliveira Rodrigues, João Vitor. *Technology as a Survival Factor*. 2025. 42 p. Undergraduate Thesis (Bachelor's Degree) – Computer Engineering Program, Institute of Technology and Leadership, São Paulo, 2025.

This work has a technical and entrepreneurial character, supporting the development and business plan of ViaFacilis, a startup focused on the management of small businesses. The project originates from the identification of a critical gap in the Brazilian ecosystem: the high mortality rate of Individual Microentrepreneurs (MEIs), mainly caused by management inefficiencies. The initiative is justified by the premise that accessible technology is a determining factor for the survival and scalability of enterprises in the current scenario. Based on this assumption, ViaFacilis aims to directly reduce this mortality rate by offering a centralized, modular, and offline-first

platform, seeking to reduce manual processes carried out using fragmented technologies.

The methodology comprised four strategic phases: validation of the problem based on SEBRAE data, legal structuring and risk analysis, technical development of the Minimum Viable Product (MVP), and the formulation of the go-to-market strategy. The results obtained from the MVP demonstrate the solution's ability to optimize cash flow, inventory control, and the generation of the DASN SIMEI, centralizing the main management demands of MEIs.

Keywords: entrepreneurship; MEI management; startup; business plan.

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List of Abbreviations and Acronyms

CAC – Customer Acquisition Cost
 ICP – Ideal Customer Profile
 MEI – Individual Microentrepreneur
 MVP – Minimum Viable Product
 MVO – Minimum Viable Offer
 PDV – Point of Sale
 SaaS – Software as a Service

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1 Introduction

The Brazilian business environment presents significant challenges for Individual Microentrepreneurs (MEIs). Statistics from SEBRAE indicate that a substantial share of these businesses—approximately 29%—cease operations within five years. Among the factors associated with this scenario are poorly structured operational management practices, often based on manual and informal controls, the lack of clarity regarding margins, profitability, and pricing, as well as difficulties in complying with tax obligations. These issues largely stem from informality in financial organization, which undermines decision-making and the sustainability of the business in the medium term.

1.1 Context and Motivation:

The area of focus of this work lies at the intersection of information technology and microenterprise management. In the Brazilian context, a relevant digital gap can be observed: although the market offers robust and well-established Enterprise Resource Planning (ERP) systems, most of these solutions do not adequately address the operational reality of Individual Microentrepreneurs (MEIs), which is characterized by low digital maturity, limited financial resources, and a lack of time for training.

Within this scenario, a significant market opportunity is identified, as more than 8 million MEIs in Brazil still manage their businesses in an analog manner, through handwritten records, or in an improvised way, using non-integrated spreadsheets and instant messaging applications. This context reveals the existence of a substantial addressable market, with a Serviceable Available Market (SAM) estimated at BRL 7.77 billion, composed of entrepreneurs who demand intermediate solutions capable of acting as a bridge between the informality of manual records and the complexity of traditional corporate software.

1.2 Problem Definition and Value Proposition:

The central problem addressed in this project can be structured into three fundamental axes that directly impact the operational routine of Individual Microentrepreneurs (MEIs):

- a) **Cash flow disorganization:** the lack of separation between personal and business finances, combined with error-prone manual recordkeeping, hinders the visualization of the actual cash balance available and compromises financial decision-making;
- b) **Lack of financial intelligence:** the absence of structured pricing criteria and limited awareness of effective profit margins per product often lead to selling at a loss or to business stagnation;
- c) **Regulatory obligations:** the fear of incorrectly completing the Annual Declaration of the Simples Nacional for Individual Microentrepreneurs (DASN-SIMEI), as well as reliance on accounting services for basic operational activities, generates additional costs and legal uncertainty.

In light of these challenges, the value proposition of the ViaFacilis system consists of developing a Point of Sale (POS) solution oriented toward visual simplicity and offline-first operation. The solution aims to mitigate the identified pain points through the following deliverables:

- **Simplified cash management:** an intuitive interface for the visual organization of financial inflows and outflows, reducing the user's learning curve;
- **Real-time profit calculation:** automated generation of reports that convert sales volume into net profit indicators and contribution margins;
- **Facilitated tax reporting:** automatic consolidation of daily records to support the preparation of the DASN-SIMEI, ensuring tax compliance and greater peace of mind for the entrepreneur.

1.3 Objectives of the Work:

1.3.1 General Objective

To create and validate an integrated computational management solution (MVP) aimed at Individual Microentrepreneurs (MEIs) and to develop a structured business plan that demonstrates its technical and economic feasibility, as well as its market entry strategy.

1.3.2 Specific Objectives

- a) To develop the Minimum Viable Product (MVP) of the ViaFacilis system (version 1.0.2), using an offline-first-oriented architecture based on Electron and React technologies, ensuring the implementation of critical point-of-sale (POS) functionalities, inventory control, and simplified financial management;
- b) To validate the developed solution by conducting usability tests with real users from the retail and service sectors, in order to verify the interface's alignment with the target audience profile and the effectiveness of the solution in mitigating financial disorganization;
- c) To define the revenue model by structuring a monetization strategy based on recurring subscriptions (Software as a Service – SaaS), with an estimated average ticket of BRL 80.00, as well as to project operational costs and determine the financial break-even point;
- d) To establish competitive differentiators grounded in the application of the Blue Ocean Strategy methodology, with the aim of positioning the product beyond competition based solely on price, through the development of solutions aligned with the operational reality of Individual Microentrepreneurs, with an emphasis on simplifying tax obligations, enabled by the use of temporal filters associated with the requirements of the annual report (DASN-SIMEI).

1.4 Justification and Contributions:

The relevance of this project is grounded in three fundamental pillars:

- a) **Market relevance:** considering a universe of more than 11 million Individual Microentrepreneurs (MEIs) in Brazil, of whom approximately 29% cease their

activities prematurely due to management failures (SEBRAE, 2024), the project addresses a latent demand for professionalization. The proposed solution seeks to fill the gap left by complex and costly enterprise management systems by offering a tool specifically designed for the digital maturity level of this audience;

b) **Technological relevance:** the project contributes by demonstrating the application of modern web technologies, such as JavaScript and React, in the development of robust desktop applications through the Electron framework. The proposed architecture addresses the challenge of unstable connectivity in Brazilian retail by ensuring local offline operation with subsequent data synchronization, combining the performance of native systems with the agility of web development;

c) **Economic relevance:** by providing tools that mitigate so-called “financial blindness” and facilitate tax compliance—especially with regard to the Annual Declaration of the Simples Nacional for Individual Microentrepreneurs (DASN-SIMEI)—the system contributes to extending the survival and growth of small businesses. Strengthening the MEI ecosystem generates direct positive impacts on the national economy through job retention and increased tax revenue.

1.5 Work Structure:

This work is organized into chapters that guide the reader from the contextualization of the problem to the analysis of the results obtained from the development and validation of the proposed solution.

Chapter 1 – Introduction presents the context of Individual Microentrepreneurs in Brazil, the motivation for the study, the definition of the research problem and the value proposition, as well as the general and specific objectives of the work. The chapter also includes the technical and academic justification of the study and a description of the adopted structure.

Chapter 2 – Solution Development constitutes the core of the work. Initially, the market premises and the problem, solution, and value hypotheses that underpin the system proposal are defined. Subsequently, market analysis and sizing are conducted, covering the Total Addressable Market (TAM), Serviceable Available

Market (SAM), and Serviceable Obtainable Market (SOM) metrics, as well as customer segmentation and persona characterization. The chapter also presents the competitive analysis and the differentiation factors of ViaFacilis in relation to existing solutions. Next, the adopted technological solution is described, including requirements, architecture, and technologies used, as well as the process of development, integration, and technical evaluation of the MVP. The chapter is complemented by the presentation of the business plan, encompassing strategic analysis, the business model, marketing and sales strategies, and financial projections. Finally, the validation procedures and the results obtained are presented, including performance indicators, identified risks, and their respective mitigation plans.

Chapter 3 – Conclusion synthesizes the main contributions of the work, evaluates the achievement of the proposed objectives, discusses the limitations identified during the development and validation of the solution, and presents recommendations for future work aimed at the evolution of the platform.

Finally, the post-textual elements gather the bibliographic references used in the research, as well as the annexes considered relevant for the understanding and support of the study.

2 Solution Development

2.1 Definition of Market Assumptions and Hypotheses:

The central hypothesis of this work is that Individual Microentrepreneurs (MEIs) face critical barriers in business management, which can be summarized into three main axes: cash flow disorganization resulting from inadequate operational practices; the absence of effective financial control, associated with a lack of awareness of actual profits and the absence of structured pricing criteria; and the complexity of complying with tax obligations, related to the fear of errors in fulfilling legal requirements.

It is assumed that this audience is willing to invest in a paid solution, provided that it is capable of reducing the complexity of traditional enterprise management systems and replacing inefficient practices based on manual records and non-integrated spreadsheets.

2.1.1 Problem Hypothesis

The central hypothesis of this work is that Individual Microentrepreneurs (MEIs) face critical barriers in business management, which can be summarized into three main axes: cash flow disorganization resulting from inadequate operational practices; the absence of effective financial control, associated with a lack of awareness of actual profits and the absence of structured pricing criteria; and the complexity of complying with tax obligations, related to the fear of errors in fulfilling legal requirements.

It is assumed that this audience is willing to invest in a paid solution, provided that it is capable of reducing the complexity of traditional enterprise management systems and replacing inefficient practices based on manual records and non-integrated spreadsheets.

2.1.2 Solution Hypothesis

It is hypothesized that a computational solution oriented toward a simplified Point of Sale (POS), with an architecture designed for offline-first operation, constitutes the most appropriate approach to addressing the identified problem. Unlike exclusively mobile applications, which are often limited in functionality, or fully cloud-based systems that depend on constant connectivity, the proposed architecture offers greater stability for the point-of-sale environment.

Furthermore, it is assumed that the solution's value delivery is not limited to mere data recording, but rather to the ability to automatically transform this information into real-time profit indicators, as well as to facilitate compliance with the Annual Declaration of the Simples Nacional for Individual Microentrepreneurs (DASN-SIMEI), acting as an automated managerial support tool for the entrepreneur.

2.1.3 Value Hypothesis

The value hypothesis assumes that a recurring revenue model based on subscriptions (Software as a Service – SaaS), with an estimated average ticket of BRL 80.00, is perceived as acceptable and affordable by the target audience. This amount represents a significantly lower cost than hiring additional administrative staff or making recurrent use of accounting consultancy services.

It is estimated that this pricing level, combined with reduced time spent on bureaucracy and increased visibility into the business's financial results, will provide a noticeable return on investment (ROI) within the first months of use, favoring the economic viability and commercial scalability of the solution, as it contributes to the generation of perceived value that is expected to translate into customer retention.

2.2 Market Sizing and Analysis

In this subsection, the results of the market size estimates obtained through the TAM, SAM, and SOM methodology are presented. The calculations are based on the premise of an average monthly ticket of BRL 80.00 (SaaS subscription model), applied to the universe of active Individual Microentrepreneurs (MEIs) in Brazil.

2.2.1 Market Size (TAM, SAM, SOM):

The market analysis reveals significant economic potential, given the widespread presence of small businesses across the country. The structuring of the market levels was defined as follows:

TAM (Total Addressable Market): This level encompasses all active MEIs in the Commerce and Services sectors in Brazil. In theory, any business within this profile could benefit from a management system.

Base: over 11.6 million MEIs.

Annual Potential: BRL 11.17 billion.

SAM (Serviceable Available Market): An operational segmentation was carried out focusing on the technological reality of the country. The SAM represents MEIs who currently use manual methods (paper, pen, notebooks) or disconnected spreadsheets for business management, constituting the audience that urgently requires digitalization.

Base: over 8.0 million MEIs (approximately 69% of the total).

Annual Potential: BRL 7.77 billion.

SOM (Serviceable Obtainable Market): This represents the realistic short- to medium-term market capture target, considering ViaFacilis's initial operational capacity and marketing strategies. The goal was defined as capturing 1% of the operational segment (SAM).

Base: over 80,000 customers.

Annual Potential: BRL 76.80 million.

Table 1 summarizes the market sizing and the projected annual revenue potential.

Table 1 – Market Sizing (TAM, SAM, SOM)

Market Level	Segment Description	Customer Base (Estimated)	Average Ticket	Annual Revenue Potential
TAM	All Commerce and Service MEIs	11,600,000	R\$ 80.00	R\$ 11.17 Billion
SAM	MEIs with manual/analog management	8,000,000	R\$ 80.00	R\$ 7.77 Billion
SOM	Capture target (1% of SAM)	80,000	R\$ 80.00	R\$ 76.80 Million

Source: Prepared by the author (2025)

The analysis of the Serviceable Obtainable Market (SOM), estimated at 80 thousand customers, indicates that the business model is economically viable, as even a conservative market penetration of approximately 1% proves sufficient to sustain operations and enable financial scalability.

2.2.2 Customer Segmentation and Profiling

In order to guide product development and the definition of communication strategies, a representative persona of the ideal customer for the ViaFacilis solution was developed. This profile was constructed based on qualitative data and the analysis of the main recurring difficulties observed in the sector.

2.2.2.1 Target Segment

The target segment comprises Individual Microentrepreneurs operating in the food sector (such as bars and snack shops), small retail, and service provision, who seek to professionalize the management of their businesses but show resistance to adopting complex management systems with a steep learning curve.

2.2.2.2 Persona: Expanding Microentrepreneur

The persona defined to represent this segment is named Raoni Santos, characterized as a microentrepreneur in the expansion phase of his business.

Demographic Profile

A 38-year-old individual, owner of a skewer restaurant (espeteria) located in the municipality of Jaguapitã, Paraná. He has completed secondary education and falls within economic class B, considering household income.

Digital Behavior

A frequent user of everyday digital tools such as instant messaging applications and search engines, demonstrating basic proficiency in digital technologies, but with limitations in the use of more complex corporate software.

Primary Motivation

To expand sales volume and professionalize business management, with the aim of

promoting sustainable growth and avoiding inconsistencies in complying with tax obligations.

Main Difficulties and Needs

The main difficulties identified are concentrated in three dimensions:

- a) **Cash flow management:** inconsistencies in the daily control of financial inflows and outflows, resulting from the use of manual and non-systematized records;
- b) **Financial control and pricing:** lack of clarity regarding the actual profit of the business and difficulties in defining appropriate product prices;
- c) **Compliance with tax obligations:** insecurity in organizing the information required to prepare the Annual Declaration of the Simples Nacional for Individual Microentrepreneurs (DASN-SIMEI), with fear of penalties arising from errors or omissions.

2.3 Competitive Analysis and Differentials:

In this subsection, an analysis of the competitive environment is presented, identifying the main players competing in the market for microenterprise management solutions, as well as the factors that differentiate the ViaFacilis value proposition. To this end, the Blue Ocean Strategy methodology is adopted, with the objective of mapping the competitors' value curve and highlighting the differentiation proposed by the solution.

2.3.1 Identification of Competitors

The market for management solutions for small businesses is characterized by a high degree of fragmentation, being polarized between more complex technological tools and informal manual control methods.

Direct competitors (entry-level ERPs): These correspond to well-established SaaS (Software as a Service) platforms such as Bling, Tiny, and MarketUp. These solutions offer comprehensive management functionalities, including inventory control, financial management, and the issuance of fiscal documents, and are primarily

targeted at companies with a higher level of digital maturity, especially in the e-commerce segment.

Indirect competitors (manual methods and spreadsheets): These simultaneously represent the greatest competitive challenge and the main market opportunity, as indicated in the SAM analysis. They include the use of notebooks, non-integrated electronic spreadsheets, and informal management practices via instant messaging applications. Their main attractions lie in their low or non-existent cost and low technological barriers, although they present a high risk of operational errors and a lack of analytical intelligence.

2.3.2 Competitor Analysis

The comparative analysis shows that, despite the functional robustness of direct competitors, these solutions present relevant limitations in addressing the simplicity needs of the typical Individual Microentrepreneur (MEI).

Pricing and Business Model: The main management and point-of-sale (POS) platforms aimed at micro and small businesses adopt different strategies. Established solutions such as Bling and Tiny predominantly operate under a monthly subscription model. Although suitable for structured companies, this fixed cost represents a significant financial barrier for MEIs at an early stage or with low cash flow. MarketUp positions itself as an entry-level solution by offering a free version; however, its monetization occurs through the sale of additional services and paid POS plans, whose final cost tends to be higher than that proposed by ViaFacilis. In contrast, ViaFacilis proposes an effective freemium model, eliminating initial financial barriers and aligning with the microentrepreneur's investment capacity.

Functionality and Complexity: The analyzed competitors tend to adopt a generalist approach, offering a broad set of features that results in dense interfaces and prolonged learning curves. The predominant functionalities are descriptive in nature, limited to the presentation of reports on past events.

Competitors' Strengths: These include platform robustness, integration with established marketplaces such as Shopee and Mercado Livre, and the ability to issue complex fiscal documents, such as the Electronic Invoice (NFe).

Competitors' Weaknesses: There is an excess of functionalities with little relevance for MEIs operating in a physical point-of-sale context, full dependence on internet connectivity in fully web-based solutions, and a lack of strategic guidance, as the systems do not direct users regarding actions to be taken based on the presented data.

2.3.3 Competitive Advantage and Differentiating Factors

ViaFacilis proposes the creation of a new market space, aligned with the Blue Ocean concept, by shifting the competitive axis from a dispute over the quantity of features to the effective generation of value and customer success.

The main differentiating factor of the solution lies in the transition from a descriptive model to a prescriptive model of management support.

Prescriptive business intelligence (suggestion engine): Unlike competing solutions, which are limited to presenting charts and reports, ViaFacilis acts as a strategic guidance system, comparable to a “Google Maps for entrepreneurship.” Based on the analysis of operational data, the system suggests practical actions, such as price adjustments based on profit margins and alerts related to tax obligations, such as the Annual Declaration of the Simples Nacional for MEIs (DASN-MEI).

Offline-first architecture: While most competing solutions require continuous internet connectivity, the Electron-based architecture adopted by ViaFacilis ensures operational stability at the point of sale, regardless of connection quality, synchronizing data only when possible.

Gamified journey: The reduction of the learning curve is promoted through a growth journey structured in missions, guiding the progressive use of the system and mitigating the technological resistance frequently observed in this audience.

In summary, ViaFacilis's competitive advantage is grounded in the combination of operational simplicity with automated intelligence, occupying the gap between traditional manual methods—characterized by ease of use but limited analytical capacity—and conventional ERP systems, recognized for their robustness but marked by high complexity of use.

2.3.4 MVP Scope Definition and Solution Evolution

At the Minimum Viable Product (MVP) development stage, the ViaFacilis solution was deliberately designed with a focus on validating operational simplicity, usability, and the feasibility of offline operation in the context of the Individual Microentrepreneur (MEI). The implementation of prescriptive intelligence features and a gamified journey was intentionally postponed at this initial stage, as these layers depend directly on the recurrence, consistency, and quality of operational data generated by the user. Thus, the MVP focuses on validating the target audience's adoption of basic management digitalization, establishing the necessary foundation for the progressive introduction of strategic guidance and engagement mechanisms in subsequent versions. This decision reflects an incremental development approach aligned with best practices in digital product validation and risk mitigation in the innovation process.

2.4 Technological Solution

This section presents the software architecture and the engineering decisions adopted in the development of ViaFacilis, with the aim of meeting performance requirements in hardware-constrained environments and ensuring full offline operation—factors considered critical for the Individual Microentrepreneur (MEI) audience.

2.4.1 Requirements and Specifications:

The requirements elicitation was conducted with a focus on minimizing computational resource consumption and eliminating dependence on continuous connectivity, taking into account the operational context of small-scale retail.

2.4.1.1 Critical Functional Requirements

The system prioritizes the **Sales Module (POS)**, allowing the registration of sales, automatic calculation of totals, and shopping cart management. **Inventory Control** includes monitoring of minimum stock levels, issuance of alerts, and an auditable record of product entries and exits. Additionally, there is a requirement for a **native installer in executable format (.exe)**, with a maximum size of 255 MB, to enable distribution through offline means, such as USB devices or instant messaging applications.

2.4.1.2 Non-Functional Requirements

Regarding non-functional requirements, criteria related to **Quality of Service (QoS)** were established as follows:

- a) **Performance:** screen loading times must not exceed two seconds, ensuring agility in the point-of-sale environment.
- b) **Efficiency:** the system must be optimized for execution on equipment with limited RAM availability, a common characteristic in small-scale retail.
- c) **Resilience:** the application must operate fully in offline mode (RNF01), with secure local data persistence and code-protection mechanisms, including obfuscation techniques.

2.4.2 Architecture and Technology:

The solution adopts a desktop architecture based on isolated processes, structured around the Main Process and Renderer Process models, orchestrated by the Electron framework. This approach enables the packaging of modern web technologies into a native application for the Windows operating system, ensuring portability and controlled access to the local file system.

2.4.2.1 Process Structure and Communication

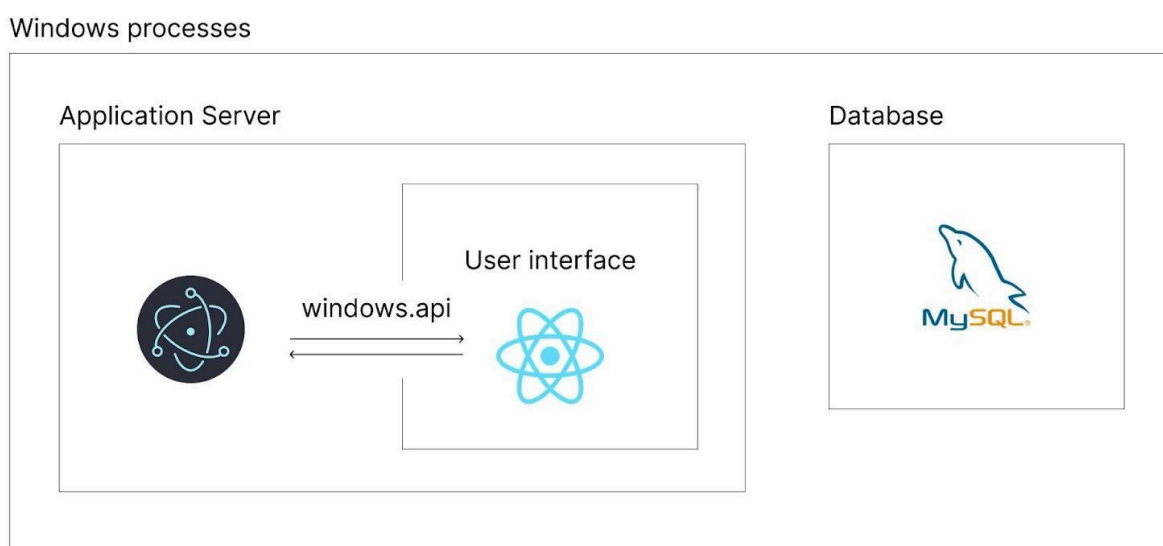
According to the system architecture diagram, the application operates through two distinct processes:

- a) User Interface (Renderer Process): developed in React, it is exclusively responsible for the presentation layer and user interactions.

b) Application Server (Main Process): acts as a local backend, centralizing business rules and access control to system resources.

Communication between these processes does not occur through conventional HTTP requests, but through a secure bridge (windows.api) based on Inter-Process Communication (IPC) mechanisms. This strategy reduces latency associated with internal communication and contributes to increased application security.

Figure 1 - Overall ViaFacilis architecture based on Electron



Source: Prepared by the author (2025)

2.4.2.2 Backend and Resource Efficiency

For logical processing in the Main Process, Node.js was chosen in conjunction with the Express.js framework. This decision is based on a comparative analysis of average RAM consumption in an idle state, as observed in different technology stacks:

- a) Express.js (Node.js): average consumption between 30 MB and 50 MB;
- b) Django (Python): average consumption between 90 MB and 150 MB;
- c) Spring Boot (Java): average consumption between 120 MB and 300 MB.

Table 2 - Technical Infrastructure Comparison

Framework	Language	Average RAM (Idle)	Technical Comment

Express.js	Node.js	30–50 MB	Flexible and easy to use; moderate consumption.
Flask	Python	40–70 MB	Lightweight, but grows with extensions; caution with scalability.
Django	Python	90–150 MB	Full framework, but heavy for low-resource environments.

Source: Prepared by the author (2025)

The choice of Express.js proved to be the most suitable for the context of modest hardware, offering a favorable balance between development productivity and low computational resource consumption.

2.4.2.3 Data Persistence Strategy

Data persistence is performed through the MySQL database management system, modularly structured to support both local POS operation and future cloud synchronization. The modeling was organized into two main domains:

- a) Central Structure (Core ERP): responsible for multi-tenancy management, access hierarchy (user_workspace_access), and functionality catalog (tools), prepared for a SaaS model scalability scenario.
- b) POS Module (Local): oriented towards maximizing transaction speed. Critical tables, such as pos_sales and pos_stock_movements, were designed to ensure relational data integrity in an offline environment and enable complete auditing of movements.

2.4.3 Development and Implementation (MVP):

The development of the Minimum Viable Product (MVP) followed the **Scrum methodology**, adopting an iterative and incremental approach organized into five biweekly sprints. The primary objective was to validate the hypotheses regarding the

main challenges faced by Individual Microentrepreneurs (MEIs) and to test critical functionalities that address their identified needs, ensuring the progressive delivery of value in alignment with the project's goals.

2.4.3.1 Backend Implementation

The MVP architecture was based on an adapted **MVC (Model-View-Controller)** pattern, using the **Electron framework** to package the POS module and the **Node.js environment with TypeScript** for implementing the business logic. The backend implementation was structured along the following axes:

- a) **Logical structure:** a clear separation was established between **Controllers**, responsible for business rules, and **Models**, responsible for data persistence in MySQL, ensuring greater organization, readability, and maintainability of the code.
- b) **Functional division:** the system was segmented into the **Local POS**, responsible for sales operations, inventory control, and report generation with offline support, and the **Central Server**, prepared for future client management and multi-tenancy support.
- c) **Security:** internal communication between processes was implemented using Electron's **ipcMain**, **ipcRenderer**, and **preload** modules, ensuring the integrity and control of message exchanges between application layers.

2.4.3.2 Frontend Implementation

The frontend was developed as the main point of user interaction with the system, prioritizing usability, visual clarity, and ease of navigation. Technologies employed included **React**, **TypeScript**, **Tailwind CSS**, and **Vite**. The interface was built following component-based design and modular folder organization practices. Implemented functionalities included product registration, complete sales workflow, operational report generation, and basic system configuration options.

2.4.3.3 Integration and Preliminary Validation

Frontend and backend integration occurred after the independent development of each module. During this phase, the MVP underwent initial installation and

operational tests to verify the correct functioning of essential workflows and the consistency of the architecture.

As a result, **ViaFacilis v1.0.2** was delivered, characterized as a functional and integrated MVP capable of meeting the initial demands of the target audience, prioritizing essential functionalities while deferring complementary implementations, such as electronic invoice issuance, to future evolutionary stages.

Details regarding technical, integration, and acceptance tests are described in **Section 2.5 – Testing and Technical Evaluation**.

2.4.4 Testing and Technical Evaluation:

The software quality assurance process was conducted through a progressive testing strategy, integrated with the iterative methodology adopted in the project. The primary objective was to validate the technical integrity of the hybrid desktop architecture and ensure the system's functional alignment with the operational needs of Individual Microentrepreneurs (MEIs).

Integration Testing focused on the communication between the business logic (Backend/Main Process) and the user interface (Frontend/Renderer Process). This phase ensured that critical requests, such as sales registration and inventory movements, were correctly interpreted and persisted in the local MySQL database. Additionally, the packaging of the application via Electron was validated, guaranteeing correct installation and execution of the executable in a Windows environment.

Functional and Usability Validation was conducted in two sessions with real users, simulating distinct scenarios: a burger restaurant (simple sales) and an açaí shop (customizable products with multiple options). Participants completed full workflows, including the registration of groups and products, sales recording (both local and

delivery), report generation, and adjustments in system settings, allowing observation of actual user interaction with the application.

The results demonstrated the operational efficiency of the system, with average execution times compatible with the expectations of the target audience: sales registration between 2 and 4 minutes, simple product registration in 3 minutes, and products with multiple variations in approximately 5 minutes. During the evaluation, opportunities for interface and workflow improvements were identified, such as enhancing visual feedback when blocking products with negative margins and adjusting table-linking navigation. These improvements were implemented during the development cycle, contributing to a clearer, faster, and more intuitive user experience.

From a technical perspective, the system demonstrated operational robustness, functioning stably and without data loss during the tested workflows, confirming the effectiveness of the modular architecture and the feasibility of offline operation.

As a result, **ViaFacilis v1.0.2** was consolidated as a functional MVP, suitable for homologation and controlled use in a real environment.

2.5 The Business Plan

The market analysis of ViaFacilis aims to demonstrate the system's commercial viability by connecting the technological solution to the real needs of Individual Microentrepreneurs (MEIs) in Brazil.

2.5.1.1 Segmentation and Target Audience (Persona)

The addressable market includes MEIs in the commerce and services sectors who seek to professionalize business management but face barriers due to the complexity of traditional tools. The segmentation prioritizes entrepreneurs with growth ambitions but operational disorganization and low digital maturity.

To represent the target audience, the persona “**Raoni Santos**” was defined:

- **Profile:** 38 years old, owner of a skewer restaurant in Jaguapitã (PR), with a household income corresponding to class B.
- **Behavior:** Practical, sales-oriented, and resistant to complex bureaucratic processes.
- **Latent Pains:** Cash flow disorganization (“Cash Chaos”), lack of awareness of actual profit (“Financial Blindness”), and fear of tax non-compliance (“Bureaucratic Fear”).
- **Need:** A solution that simplifies daily operations and allows the entrepreneur to focus on revenue-generating activities.

2.5.1.2 SWOT Analysis (Strengths, Weaknesses, Opportunities, and Threats)

The strategic analysis of the business environment highlights both potential and risks:

- **Strengths:** The offline-first architecture ensures stability in locations with limited connectivity. The simplified interface and prescriptive intelligence (“Suggestion Engine”) reduce the learning curve, aligning technology with user experience.
- **Weaknesses:** The initial bootstrap operation limits resources for marketing and support team expansion. The brand is still unknown and needs to build trust with the target audience.
- **Opportunities:** Over 8 million MEIs still rely on manual methods, representing potential for digitalization. The complexity of Brazilian taxation generates ongoing demand for solutions that simplify fiscal obligations, such as DASN-SIMEI.
- **Threats:** Established competitors may lower prices or launch simplified versions of their ERPs. Cultural resistance to changing habits requires continuous user education and engagement efforts.

2.5.1.3 Competition and Competitive Differentiation

As detailed in Section 2.3.2, competitors offer robust solutions but present significant barriers for the target audience, such as complex interfaces, subscription costs, and reliance on constant internet connectivity. The use of manual methods (notebooks and spreadsheets) represents indirect competition, maintaining gaps in financial visibility and lacking prescriptive intelligence.

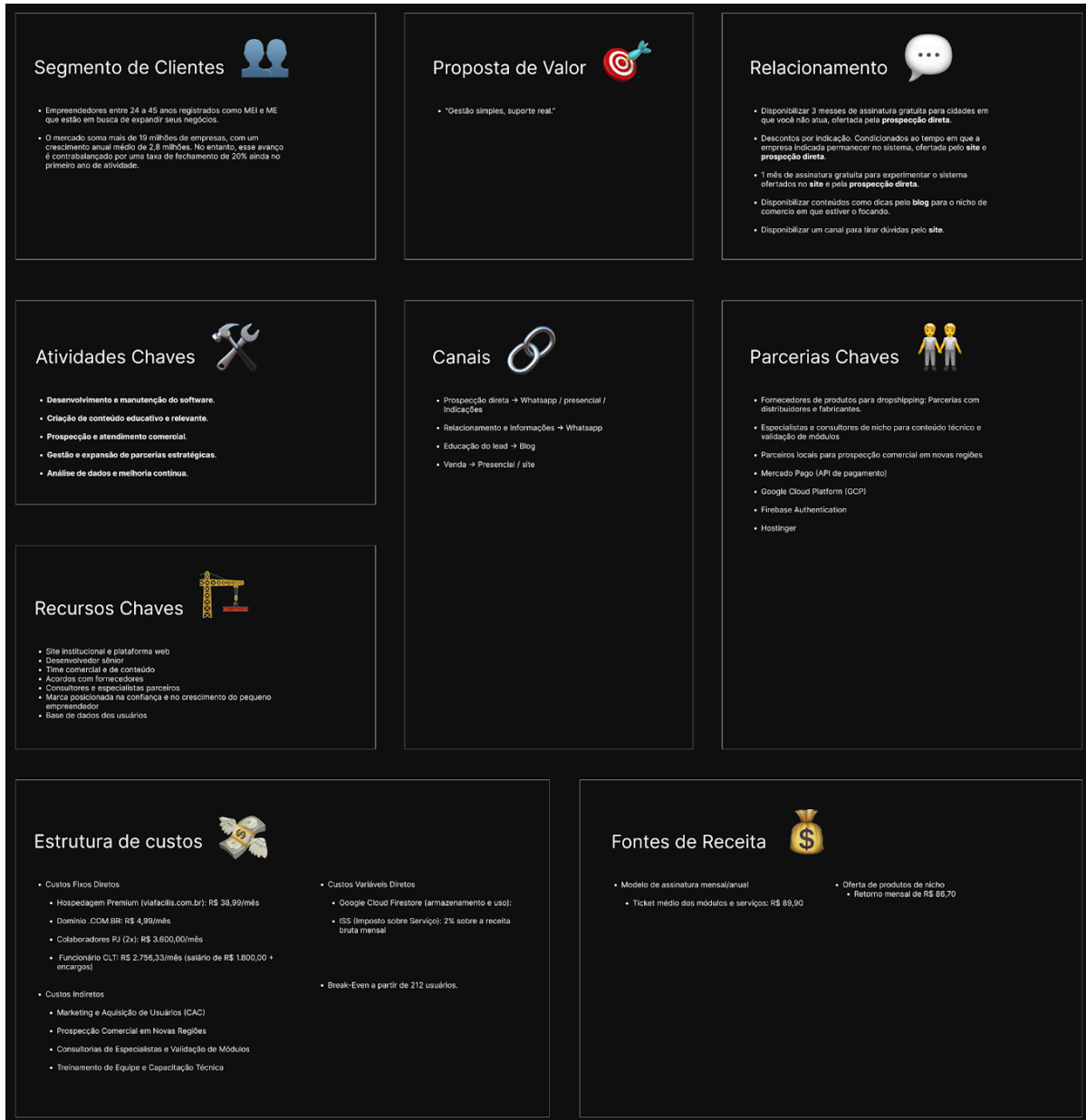
The strategic positioning of ViaFacilis seeks to explore a new value space by offering:

- **Prescriptive Business Intelligence:** Suggests future actions based on business data, acting as a digital consultant.
- **Focus on MEI Reality:** Aligning the solution with the operational reality and needs of Individual Microentrepreneurs ensures that features, interface, and communication are adapted to the target audience's profile.
- **Technological Resilience:** Complete offline operation ensures greater stability compared to purely web-based solutions.

2.5.1 Business Model (Business Model Canvas - BMC):

The ViaFacilis business model was structured using the **Business Model Canvas** methodology, providing an integrated view of how the solution creates, delivers, and captures value. The core logic is based on a **modular and scalable SaaS operation**, specifically targeted at Individual Microentrepreneurs (MEIs) in the commerce and services sectors.

Figure 2. Business Model Canvas



Source: Author (2025)

Customer Segments

The target audience consists of Individual Microentrepreneurs (MEIs) operating in retail, food services, and general service sectors, who seek to professionalize the management of their businesses but have low digital maturity and are averse to complex software. The representative persona, “**Raoni Santos,**” exemplifies this client profile: a practical entrepreneur, sales-focused, growth-oriented, and in need of simplified tools.

Value Propositions

ViaFacilis' value proposition addresses the main pains of MEIs:

- **Fear-Free Management:** Simplified financial control that translates sales into real-time net profit, combating “Financial Blindness.”
- **Operational Resilience:** Full offline POS functionality (Hybrid Desktop Architecture), ensuring continuity of operations even without internet.
- **Facilitated Compliance:** Automated organization of data for the Annual Declaration (DASN-SIMEI), reducing the risk of penalties and increasing user confidence.

Channels

Distribution and communication with MEIs occur through low-friction channels:

- **Direct Distribution:** Proprietary website for self-service installer download (.exe).
- **Communication:** WhatsApp for support and engagement, aligned with the persona's preferred channel.
- **Partnerships:** Digital accounting offices act as qualified referral channels.

Customer Relationships

The relationship strategy prioritizes scalability and automation:

- **Self-Service and Automation:** Initial installation and configuration performed by the user.
- **Growth Journey:** Internal engagement mechanisms and gamification encourage full adoption, increasing **Lifetime Value (LTV)**.

Revenue Streams

The monetization model is based on recurring revenue:

- **SaaS Subscription:** Monthly fee with an average ticket of R\$ 80, considered affordable for MEIs.
- **Freemium Model:** A functional free version to acquire users, with the possibility of upgrading to paid plans as needed.

Key Resources

Essential resources include:

- **Technological:** Proprietary source code (Electron, React, Node.js) and modular database.
- **Human:** Development team for ongoing maintenance and technical support tailored to MEIs.

Key Activities

Core activities required to operate the model include:

- Product development and maintenance.
- Technical support and customer service.
- Digital marketing strategies and production of educational content to acquire new users.

Key Partnerships

Strategic partnerships involve:

- **Payment Platforms:** Integration with gateways (e.g., Stripe) for automated subscription management.

- **Accountants and Consultants:** Partners who validate fiscal compliance and refer the solution to MEI clients.

Cost Structure

The financial structure was designed to be lean and scalable:

- **Fixed Costs:** Personnel (development and support), pro-labore, and basic infrastructure.
- **Variable Costs:** Marketing (Customer Acquisition Cost – CAC), commissions, and cloud usage proportional to the number of users.
- **Economies of Scale:** As the user base grows, fixed costs are diluted, increasing contribution margin and sustainability of the model.

2.5.2 Marketing and Sales Strategy:

The commercial strategy of **ViaFacilis** was designed to meet the needs of Individual Microentrepreneurs (MEIs), prioritizing validation of customer perceived value, ease of adoption, and organic growth in this initial phase.

Pilot Launch

The initial launch of ViaFacilis was planned as a validation program restricted to a controlled group of **early adopter users**, selected based on representativeness of the target audience. The **Minimum Viable Offering (MVO)** includes the Point of Sale (POS) module and complementary services aimed at improving the user experience, including priority support via WhatsApp, participation in defining the product roadmap, and assistance in organizing data for the Annual Declaration (DASN-SIMEI).

The price applied during this validation phase was set at **R\$ 40.00/month**, corresponding to 50% of the projected R\$ 80.00/month, with the objective of assessing the target audience's willingness to pay. The effectiveness of the strategy will be monitored through quantitative metrics, such as the number of active users, MVO adoption rate, frequency of use of the offered features, and feedback on support requests, allowing adjustments to the product and offering before expanding to a broader audience.

Communication and Positioning

Communication was designed to resonate with the MEI reality, emphasizing practical and emotional benefits: **"Fiscal Peace of Mind."** The narrative follows three pillars: identification of the pain (cash flow disorganization and anxiety regarding DASN), presentation of the solution (POS and real-time sales management), and call to action (limited availability and discount). The tone is direct, empathetic, and free of jargon, ensuring relevance and practicality are perceived by the target audience.

Customer Acquisition and Retention

Acquisition focuses on a central digital channel (**landing page**), testing two flows:

- (i) **Consultative**, with qualification via WhatsApp, and
- (ii) **Self-service**, with direct checkout.

Retention is incorporated into the user experience through gamification in onboarding, usage metrics tracking, and accessible technical support, aiming to reduce churn and increase **Lifetime Value (LTV)**.

2.5.3 Financial Projection and Feasibility:

The economic viability of ViaFacilis is based on a low fixed-cost operation (asset-light) and a recurring revenue model with high scalability potential. The following analysis details the pricing strategy, cost structure, and business sustainability indicators.

2.5.3.1 Monetization and Pricing Strategy

The business model is based on Software as a Service (SaaS). The pricing structure was designed to facilitate market entry (low friction) and ensure long-term margin expansion:

- **Validation Batch (Early Adopters):** During the pilot phase, the promotional value is R\$ 40.00/month. The goal is to accelerate Product-Market Fit and reduce the Sales Cycle.
- **List Price (Sustainability):** Established at R\$ 80.00/month. This value positions ViaFacilis as a high cost-benefit alternative to complex ERPs, maintaining a healthy contribution margin for support and R&D.

2.5.3.2 Operational Cost Structure (OPEX)

The projection below considers a scale scenario with 1,000 active users, reflecting the efficiency of cloud infrastructure and the optimization of human resources.

Table 3 – Projection of Monthly Expenses (Scenario: 1,000 Users)

Category	Description	Estimated Cost (R\$)
Human Resources	Operational and commercial support (Payroll + Taxes)	R\$ 2,734.91
Management	Founding Partners' Salary (Pró-labore)	R\$ 3,600.00
Technology (Cloud)	Hosting, DB, and API (On-demand scalability)	R\$ 62.00
Marketing & Growth	Investment in paid traffic and acquisition (CAC)	R\$ 9,000.00
Support Services	Management software, CRM, and legal consultancy	R\$ 1,750.00

Logistics/Adm	General commercial and administrative expenses	R\$ 1,500.00
TOTAL	Monthly Operational Cost (OPEX)	R\$ 18,647.09

Source: Prepared by the author (2025)

2.5.3.3 Break-even Point Analysis

For the viability calculation, the Net Revenue per user (deducting taxes and checkout fees) is considered, estimated at R\$ 78.40.

Viability Conclusion:

The break-even point is reached with only 238 subscribers, which represents just 0.3% of the Serviceable Obtainable Market (SOM) of 80,000 customers. This low market penetration requirement to achieve sustainability demonstrates the resilience of the business model and the low operational risk.

2.5.3.4 Initial Investment and Bootstrapping

The capital required for the MVP launch and the first months of operation was estimated at R\$ 9,000.00. This amount covers the direct costs of tools and initial marketing. Technical development is being executed under a bootstrapping regime, where the opportunity cost of the founders' man-hours is reinvested directly into building the company's intellectual asset.

2.6 Validation and Results

This section presents the **validation results of the project** based on empirical evidence obtained during usability tests and the real-world use of the solution by initial users. Unlike a formal market launch experiment, the validation presented is limited to the scope defined for this **Undergraduate Thesis**, focusing on the technical validation of the product and the initial observation of adherence to the business model, based on early adopter users already discussed in Section 2.4.4.

2.6.1 Validation Methodology:

The validation strategy adopted in this work was conducted on two complementary fronts, within the limits of the academic project scope.

The first front corresponds to the technical validation of the Minimum Viable Product (MVP), focusing on usability and the functioning of essential features. For this purpose, usability tests were carried out in controlled scenarios, as described in Section 2.4.4, involving real users representing the defined target audience (Microentrepreneurs).

Two usage scenarios were considered: rapid sales operations with simple products and operations with customizable products, including multiple add-ons and variations. Users performed critical system tasks, such as product registration, sales at the counter and through delivery service, access to reports, and general settings. During these activities, metrics related to task execution time and the occurrence of errors were observed.

The second validation front refers to the initial observation of solution acceptance in a real-use context. Two early adopter users, previously introduced in Section 2.4.4, began using the system in a productive environment after technical testing. These users were onboarded through direct marketing, without a structured launch campaign or formal market experiment, reinforcing the observational nature of this validation.

2.6.2 Market Validation Results:

The data obtained from usability tests and continuous system use by initial users provided relevant **quantitative and qualitative information** for product evaluation.

Regarding **operational efficiency**, the average time to register sales remained compatible with the pace of physical retail, varying according to the type of operation. Simple product registration was faster, while more complex products required additional time, highlighting opportunities for interface improvements.

The **incidence of errors** during system use was low. One issue identified was a communication gap in the interface related to blocking product registrations with negative margins, which lacked adequate visual feedback for the user. This limitation was considered significant from a user experience perspective.

As a direct result of these observations, **product adjustments** were implemented, including visual explanatory messages for blocked operations and prioritization of features aimed at speeding up product registration, such as item duplication. These decisions reflect incremental product refinements arising from technical validation and real-world use.

Concerning **business model acceptance**, both initial users converted to active customers, opting to pay the full subscription of R\$ 80.00 per month. Users continue to use the system in their daily operations, indicating initial adherence to the value proposition, although, within the scope of this work, no statistically generalizable conclusions about the broader market can be drawn.

2.6.3 Key Performance Indicators (KPIs):

Given the exploratory and applied nature of this work, **monitoring indicators** were defined with a focus on the future evolution of the product, even though not all were fully measured during the TCC period.

Key indicators highlighted include **average task execution time**, **stability of continuous use by initial users**, and **retention of active customers over time**. Metrics such as **customer acquisition cost (CAC)** and **churn rate** were identified as relevant for future project phases but were not central metrics in this study due to the absence of a formal market launch within the academic scope.

2.6.4 Risks and Mitigation Plan:

The viability and sustainability of the ViaFacilis project depend on the systematic identification of risks associated with the venture and the establishment of appropriate mitigation strategies. Considering the system's development stage, characterized as a Minimum Viable Product (MVP), as well as the target audience composed of Individual Microentrepreneurs (MEIs), the risks were analyzed and organized into four categories: **market and competitive, technological, financial, and legal and regulatory.**

2.6.4.1 Market and Competitive Risks

Risk identification:

The target audience exhibits resistance to adopting digital solutions from unestablished companies, linked to low initial trust and a high perception of risk in digital environments, which raises the system's entry barrier. Established competitors may react by reducing prices or launching simplified versions of their products.

Mitigation strategies:

ViaFacilis' communication was designed with clear language aligned with the MEI's operational reality, reducing perceived technological complexity. A hybrid acquisition model was adopted, allowing direct interaction with users to build trust. The strategic positioning focuses on differentiation through functional simplicity and offline operation—features largely unexplored by predominantly web-based competitors.

2.6.4.2 Technological Risks

Risk identification:

The MVP carries risks related to potential bugs, functional limitations, or usability adjustments that could negatively affect the user experience. Additionally, certain functionalities depend on connectivity, which may be unstable in some regions.

Mitigation strategies:

The product was framed within a pilot program, involving early users in the system's evolution, aligning expectations with the MVP's maturity. A hybrid desktop architecture developed with Electron ensures full offline operation for essential point-of-sale functions, reducing dependency on connectivity.

2.6.4.3 Financial Risks

Risk identification:

The initial self-financed operation imposes cash flow restrictions, creating liquidity risk. Some of the target audience may experience difficulties using digital payment platforms, negatively impacting conversion rates and increasing customer acquisition costs.

Mitigation strategies:

The cost structure was kept lean, prioritizing variable costs proportional to user base growth, and the breakeven point was projected at a low level. Comparative tests between payment flows were conducted to identify the approach with the lowest resistance and best financial performance.

2.6.4.4 Legal and Regulatory Risks**Risk identification:**

Changes in Simples Nacional legislation or DASN-SIMEI rules may render implemented system rules obsolete, posing a compliance risk.

Mitigation strategies:

The modular architecture allows isolated updates of fiscal rules without structural modifications to the software core. Partnerships with professionals and accounting firms ensure continuous monitoring of regulatory changes, enhancing compliance with current legal requirements.

3 Conclusion

The objectives proposed in this work were partially validated through the development and implementation of the ViaFacilis Minimum Viable Product (MVP), version 1.0.2. The solution demonstrated technical feasibility, featuring a robust architecture based on Electron and React, capable of offline operation and meeting performance requirements on devices with limited hardware.

From a business perspective, the financial analysis indicates that the adopted model has potential for sustainability, considering an estimated breakeven point of 238

users and a Serviceable Available Market (SAM) valued at R\$ 7.77 billion, according to the assumptions used in the financial model. This projection suggests that even with conservative market penetration, it is possible to achieve economic scalability of the system.

User validation demonstrated adherence to the value proposition, highlighting the simplicity of the interface, organization of cash flow, generation of real-time profit indicators, and support for fiscal compliance, addressing the identified needs of Individual Microentrepreneurs (MEIs). These results suggest that technology adapted to the MEI operational reality can contribute to professionalization and strengthen business management.

For future work, incremental evolution of the system is recommended, including the implementation of a fiscal note issuance module and the expansion of cloud synchronization functionalities. These improvements could increase the system's utility and generate additional data to validate the financial model and value hypotheses.

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Annexes

APPENDIX A – LEI COMPLEMENTAR Nº 123, DE 14 DE DEZEMBRO DE 2006

(Supporting document not prepared by the author. Establishes the National Statute for Micro and Small Enterprises, serving as the legal basis for the classification of the MEI target audience).

APPENDIX B – TABLE OF SIMPLES NACIONAL LIMITS AND RATES

(Supporting document extracted from the Brazilian Federal Revenue Service, used for reference regarding revenue brackets and tax obligations relevant to the system's financial module).